Priceora Architecture Summary

AI-Assisted Price Intelligence and Bid Costing System | Codebase handover summary | 2 November 2026

1 Application Overview

Priceora is a web application for estimating the cost of a particular task using current and historical supplier prices. Users upload quotations, invoices, purchase orders, emails, spreadsheets, PDFs, images, and supplier comparison documents. The system extracts price rows, validates uncertain data through Review, stores approved prices for search, and combines selected items and charges into a bid costing estimate.

2 Technology Stack

Layer	Confirmed implementation
Backend	Django 6.x project with Django REST Framework and Python
Frontend	React 19 SPA built with Vite, Tailwind CSS, React Router, Axios, and Recharts
Database	Microsoft SQL Server through mssql-django and ODBC Driver 17
Authentication	Django custom PriceUser model, token authentication, custom email/username backend, approval fields, and last_login
Major packages	pdfplumber, pypdf, pypdfium2, Pillow, pytesseract, extract-msg, openpyxl, pandas, rarfile, rapidfuzz, google-genai, django-cors-headers

3 High Level Architecture

SPA + REST API architecture. The React/Vite frontend runs separately from the Django REST API. The browser calls authenticated API routes; Django views/services validate and authorize requests, read/write SQL Server through models, process uploaded files, and return JSON. Frontend assets come from the Vite build; Django serves backend/static and configured media assets.

4 Main Modules

Module	Purpose
Authentication and User Management	Registration, approval/rejection, login, logout, password reset/change, profiles, roles, and administration.
Data Upload and Extraction	Upload files, parse documents, expand ZIP/RAR members, detect duplicates, and track progress.
Review and Import	Correct extracted rows, resolve supplier/date/currency/unit-price issues, and import validated rows.
Import History and Purchase Downloads	View extraction/import statuses, source files, downloaded quote files, and automatic extraction.
Product Search and Price Intelligence	Search approved price records, suppliers, currencies, source documents, and selectable products.
Bid Costing	Select products, calculate subtotal, freight, charges, tax, grand total, save drafts, extend costings, and generate PDF.
Dashboard and Usage Analytics	Show application counts, review counts, user activity, module usage, and monthly extracted-file trends.

5 Data Flow

User action → React page/component → Axios request → Django REST route/view/service → Django model query or extraction/calculation service → SQL Server and file/media processing → JSON response → frontend table, review form, progress indicator, or costing result. Typical extraction is upload → duplicate check → archive expansion → parser/OCR/AI extraction → normalization and validation → Review when uncertain → approved import → PriceHistory/Product Search → Bid Costing.

6 Access Control

Role	Confirmed access
Admin	All modules, upload/history/downloads, review, product search, bid costing, user management, analytics, and administration.
Manager	Dashboard, Needs Review, Product Search, Bid Costing, and permitted user-management operations; admin-only areas are restricted.
User	Dashboard, Needs Review, Product Search, and Bid Costing; no user management or Usage Analytics.
Scope rules	Team/department scope is not confirmed in the current Priceora codebase. Frontend route protection is supplemented by backend authentication/role checks.

7 Deployment / Runtime

- Development backend: python manage.py runserver. Production WSGI/ASGI hosting is not confirmed in the current codebase.
- Frontend: npm run dev for development and npm run build for the Vite production bundle.
- Important configuration: manage.py, costingProject/settings.py, costingProject/urls.py, api/urls.py, .env, .env.example, requirements.txt, frontend/package.json.
- Dependencies: SQL Server, ODBC Driver 17, Python packages, Node.js/npm, and 7-Zip/UNRAR when RAR extraction is required.

8 Key Design Notes

- Validation and Review handle required fields, duplicate files, duplicate currencies, supplier mapping, dates, unit prices, and uncertain OCR/handwriting.
- Import history and created/updated timestamps support reporting; Usage Analytics also uses PriceUser.last_login. Passive page views are not tracked.
- Bid costing requires at least one item and calculates item subtotal, freight, other charges, tax, and grand total from approved price records.
- Department/team analytics and detailed login/audit history are not confirmed. Archive extraction depends on server tool installation and permissions.